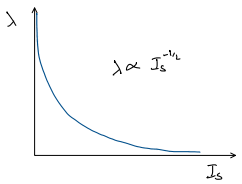
Problem 2.1

Debye length : $\lambda = \sqrt{\frac{\varepsilon RT}{F^2 \sum_j z_j^2 c_{j\infty}}}$; $I_s = \frac{1}{2}\sum_j z_j^2 c_{j\infty}$ (ionic strength) $\therefore \quad \lambda = \sqrt{\frac{\varepsilon RT}{F^2 \cdot 2I_s}}$

Net charge falling exponentially over distances on order of Debye length → i.e. distances over which electroneutrality breaks down

$\lambda = \sqrt{\frac{\varepsilon RT}{2F^2}} \cdot I_s^{-1/2}$

$\lambda \propto I_s^{-1/2}$



@ pH 7, pure water, $I_s = 1\times10^{-3}$ M ; assuming room temperature of 20°C

$K = \frac{\varepsilon}{\varepsilon_0}$; $\varepsilon_{H_2O} = 78\cdot(8.854\times10^{-14}\ \frac{F}{cm}) = 78\times(8.85\times10^{-12}\ \frac{F}{m})$

$\therefore \quad \lambda = \left(\frac{78\times8.854\times10^{-12}\times 8.314\times 293.15}{2\times(96485)^2}\right)^{1/2}\cdot\left(1\times10^{-3}\times10^{3}\right)^{-1/2}$

$\boxed{\lambda = 9.5\times10^{-9}\ m}$ ≈ 9.5 nm

+ Really small value ✓
+ Significance → Debye length just provides a general idea of the diffuse layer ↓

A really small value would mean that charge neutrality can be only broken over nm scale; As the conc. of electrolyte (or I_s ↑) increases. (This might have significance when dealing w/ double layer analysis)

@ same conditions but $I_s = 1M$: $\boxed{\lambda = 3.01\times10^{-11}\ m}$ ≈ 0.03 nm

As I_s↑, λ↓ ✓

Problem 2.2

↳ Conductivity of an electrolyte : $K = F^2 \sum z_j^2 y_j c_j$; $y_j = \frac{D_j}{RT}$ (D_j: Diffusion coefficient; y_j: Ionic mobility)

[K] = $\Omega^{-1}cm^{-1}$; c_j: ionic conc.

Water dissociates into H^+ (H_3O^+) and OH^- → @ pH 7 10^{-7} mol of H_3O^+ & OH^- ions

$K = F^2\left[(+1)^2\cdot 10^{-7}\cdot\frac{9.310\times10^{-9}\times10^3}{(8.314)(293.15)} + \frac{(-1)^2\cdot10^{-7}\cdot5.270\times10^{-9}\cdot10^3}{(8.314)\cdot(293.15)}\right]$

$K = 5.57\times10^{-9}\ \Omega^{-1}m^{-1}$ ≈ $\boxed{0.056\ mS/cm}$

↳ Sea water : Na^+, Cl^- ; $I_s = \frac{1}{2}\sum z_j^2\cdot c_{j\infty} = 0.5M$; @ room temp

$K = F^2\cdot\left((+1)^2\cdot\frac{D_{Na^+}\cdot c_{Na^+}}{RT} + (-1)^2\frac{D_{Cl^-}c_{Cl^-}}{RT}\right)$; Assuming only Na^+, Cl^- in seawater

If we assume $D_{Na^+} \approx D_{Cl^-} \approx 10^{-9}\frac{m^2}{s}$; $D_{Na^+} = 1.33\times10^{-9}\ m^2/s$; $D_{Cl^-} = 2.03\times10^{-9}\ m^2/s$ (Somewhat valid assumption)

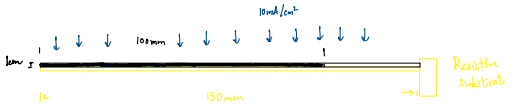
$K = F^2\cdot\frac{D_{Na^+}}{RT}\underbrace{\left((+1)^2\cdot c_{Na^+} + (-1)^2 c_{Cl^-}\right)}_{2\cdot I_s} = \frac{(96485)^2\cdot(1.0\times10^{-9})}{(8.314)(293.15)}(0.5)\cdot 2$

$= 38.2\times10^{-3}\ \Omega^{-1}m^{-1}$ or $\boxed{38\ mS/cm}$ (1 S = 1 ohm^{-1})

0.5 M NaCl → 0.0479 S/cm ≈ 47.9 mS/cm
@ Engineering Toolbox

Our estimate is a factor of 0.79 of the actual value.

Problem 2.9



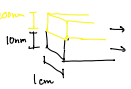
Substrate → Ru (resistive)

(a) Substrate : $R = \frac{\rho L}{A}$ (ρ: resistivity; Resistance is proportional to the no. of e⁻ that can pass through a cross section of wire)

$$R = \frac{(71\times10^{-9})(150\times10^{-3})}{(10\times10^{-9})(1\times10^{-2})}$$

$$\boxed{R = 106.5\ \Omega}$$

(b)



$$R = \frac{\rho L}{A}$$

$$R = \frac{(17\times10^{-9})(150\times10^{-3})}{(200\times10^{-9})(1\times10^{-2})}$$

$$\boxed{R = 1.3\ \Omega}$$

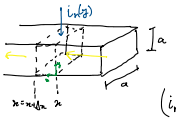
Ideally we would have to consider the resistivity of both Cu & Ru, but because Cu is much more conductive (and thicker) we only need to consider this layer

$\rho_{Cu} = 17\ n\Omega m$

(c) Constant deposition rate = $10\ mA\ cm^{-2}$; Ohms Law: $V = IR$

(current density (i))

Shell balance:



m = metal
s = substrate ; $V_m = \rho i_m L$

$$(i_m(x) - i_m(x+\Delta x))a^2 = -i_n \Delta x^2$$

$$a^2 \frac{di_m}{dx^2} = i_n \quad ; \quad i_m = \frac{V_m}{\rho_s L_s}$$

$$\frac{a^2}{\rho_s L_s}\frac{d^2V_m}{dx^2} = i_n \quad ; \quad a^2 = L_s \cdot \lambda_s$$

$$\boxed{\frac{\lambda_s}{\rho_s}\left(\frac{d^2V_m}{dx^2}\right) = i_n}$$

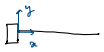
(d) Solve differential equation:

$$\frac{d^2V_m}{dx^2} = \frac{\rho_s \cdot i_n}{\lambda_s} \neq f(x)$$

$$\frac{dV_m}{dx} = \frac{\rho_s \cdot i_n}{\lambda_s}x + c$$

$$V_m = \frac{\rho_s i_n}{2\lambda_s}x^2 + cx + d$$

$$\therefore V_m = \frac{\rho_s i_n L^2}{\lambda_s 2}$$



BC:

1) @ $x=0$, $V_m = 0$ (potential at current collector is zero)

2) @ $x=L$, $\frac{dV_m}{dx} = 0$ (electronic current at end is zero)

For 10nm Ru :

$$V_m = \frac{(71\times10^{-9})(100)(150\times10^{-3})^2}{2\cdot 10\times10^{-9}} = 8V$$

For 200nm Cu :

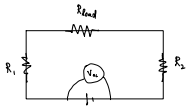
$$V_m = \frac{(17\times10^{-9})(100)(150\times10^{-3})^2}{2\cdot 200\times10^{-9}} = 0.095\ V$$

Although current flow majorly happens via Cu

↓

So Cu potential is insignificant

Problem 2.13



+ Simple, linear circuit
+ $R_{load} \to \infty$, $V_{oc} = 1V$ ← Open circuit (We assume that this is the voltage output if the cell and is constant)

(a) $R_1 = R_2 = 1\,\Omega$; $P_{load} = V_{load} \cdot I = I^2 R_{load}$.

$V_{load} = I \cdot R_{load}$

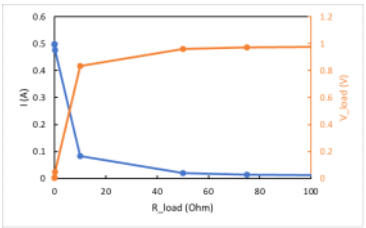
Linear/Series circuit: $R_{total} = \underbrace{R_1 \| R_2}_{2\,\Omega} + R_{load}$; $I = \dfrac{V_{oc}}{(R_{total})}$

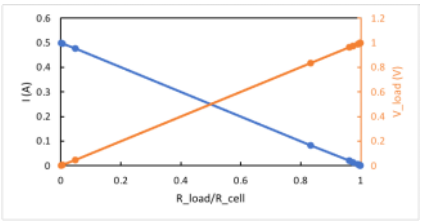
$V_{total} = V_1 + V_2 + V_{load}$

Bounds of I: @ $R_{load} = 1000\,\Omega$, $V_{load} = 1V$ (Very high R) } Open circuit

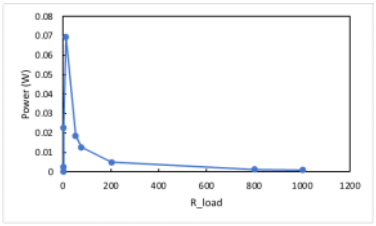
@ $R_{load} = 0.001\,\Omega$, $V_{load} = 0.01\,V$ (Very low R) } Short circuit

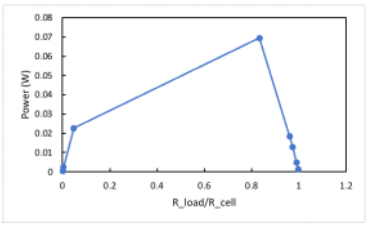
Plotting using discretized values :





(b)





* Graphs show the necessary trend: Max power occurs at a specific R_{load} well below that of V_{oc} value for R

(c) Current at which P_{max}? $P_{load} = V_{load} \cdot I$; $V_{load} = I\,R_{load}$

$P_{load} = I^2 \cdot R_{load}$; $P_{load} = \dfrac{V_{load}^2}{R_{load}}$

$P_{load} = \dfrac{V_{oc}}{(2+R_{load})^2} \cdot R_{load}$;

$$\left.\frac{\partial P_{load}}{\partial R_{load}}\right|_{V_{oc}} = V_{oc}\left[\frac{(2+R_{load})^2 - R_{load}\cdot 2\cdot(2+R_{load})}{(2+R_{load})^4}\right] = \frac{V_{oc}}{(2+R_{load})^4}\left(4 + R_{load}^2 + 4R_{load} - 4R_{load} - 2R_{load}^2\right) = \frac{V_{oc}(4 - R_{load}^2)}{(2+R_{load})^4} = \frac{(2+R_{load})(2-R_{load})}{(2+R_{load})^{\cancel{4}3}} = \frac{(2-R_{load})V_{oc}}{(2+R_{load})^3}$$

$P_{max} \Rightarrow \dfrac{\partial P_{load}}{\partial R_{load}} = 0 \Rightarrow \dfrac{(2-R_{load})V_{oc}}{(2+R_{load})^3} = 0$

$R_{load} = 2\,\Omega$ @ P_{max}

$\therefore I_{max} = \dfrac{V_{oc}}{(2+R_{load}^{max})} = \boxed{\dfrac{V_{oc} = I_{max}}{4}}$

Problem 3.1

Given $k\,[s^{-1}]$ & $T\,[°C]$

Arrhenius relationship : $k = A_0 \exp\left(-\dfrac{E_A}{RT}\right)$ (pre-exp. factor)

Linearized ⟶ $\ln k = \underbrace{\ln(A_0)}_{\text{y-intercept}} - \underbrace{\dfrac{E_A}{R}}_{\text{gradient}}\left(\dfrac{1}{T}\right)$

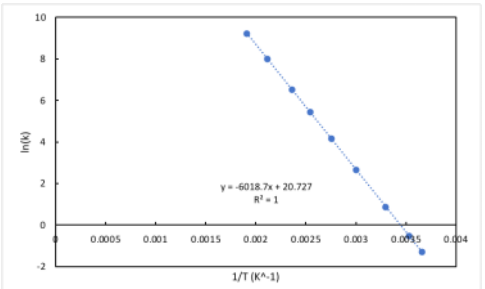
Plotting $\ln k$ vs. $\dfrac{1}{T}$ gives:

$m = -6018.7$

$c = 20.727$

$\Rightarrow -E_a = -6018.7\,(R)$

$\boxed{E_a = 50039.5\,J}$



(Assuming the last point is a typo and should be 10,000)

To achieve 10x reaction rate increase (i.e. increasing rate constant) [$r \propto k$]: (Room T = 20°C)

$$\frac{k_{new}}{k_o} = 10 \Rightarrow \frac{A_o \exp\left(\frac{E_a}{RT_{new}}\right)}{A_o \exp\left(\frac{E_a}{RT_{room}}\right)} = 10$$

$$\exp\left(-\frac{E_a}{RT_{new}} + \frac{E_a}{RT_{room}}\right) = 10$$

$$\frac{E_a}{R}\left(\frac{1}{T_{room}} - \frac{1}{T_{new}}\right) = \ln 10$$

$$\frac{1}{T_{new}} = \frac{1}{T_{room}} - \frac{R \ln 10}{E_a}$$

$$= \frac{1}{293.15} - \frac{8.314 \ln(10)}{50039.5}$$

$$T_{new} = 330.18 \text{ K}$$

$$\approx 57^\circ C$$

i.e. T increase by 37°C to achieve 10x rate increase from 20°C

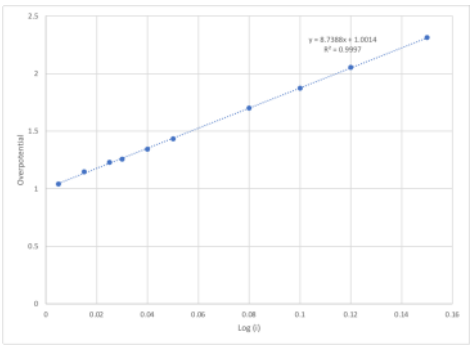
Problem 3.2

Tafel plot: Large overpotential ($\eta \gg 0$) $\Rightarrow$ $i = i_o \exp\left(\frac{\alpha_a F \eta}{RT}\right)$ ↓ Linearize

$$\log_{10} i = \underbrace{\log_{10} i_o}_{c} + \underbrace{\left(\frac{\alpha_a F \log_{10}(e)}{RT}\right)}_{m} \eta$$

(y) (x)

Plotting $\log_{10} i$ vs. η : (Inverse Tafel plot)



$m = 8.7388$ @ room temp 293.15 K

$$\frac{\alpha_a \cdot F}{RT} \cdot \log_{10}(e) = 8.7388 \quad ; \quad \log_{10} i_o = 1.0014$$

$$\alpha_a = \frac{(8.7388)\cdot(8.314)(293.15)}{(96485)(0.4343)}$$

$$\boxed{\alpha_a = 0.508}$$

$$\boxed{i_o = 10.03 \ \frac{A}{m^2}}$$

$$\frac{i_{new}}{i_{old}} = 10 \Rightarrow \frac{i_o \exp\left(\frac{\alpha_a F \eta_{new}}{RT}\right)}{i_o \exp\left(\frac{\alpha_a F}{RT}\eta_{old}\right)} = 10$$

Similar analysis as previous part: $\frac{\alpha_a F}{RT}\left(\eta_{new} - \eta_{old}\right) = \ln 10$ @ room temp: 293.15 K

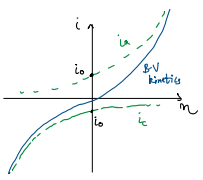
$$\boxed{\Delta\eta = 1.003 \text{ V}}$$

The overpotential increase of 1V seems a lot easier/economic to do than a ΔT increase of 37°C

Problem 7

B-V kinetics ; $\alpha_a = 0.5$; $\eta = ?$ @ 90% of total current is attributed to the anodic reaction → This means that anodic process is dominating & $0.9 = \frac{i_a}{i_o}$?

$$i = i_o\left(\underbrace{\exp\left(\frac{\alpha_a F \eta}{RT}\right)}_{\text{Anodic}} - \underbrace{\exp\left(-\frac{\alpha_c F \eta}{RT}\right)}_{\text{Cathodic}}\right)$$



i_o means the equilibrium case where anodic & cathodic reactions are happening at same rate

Since we are in the anodic regime;

$$i = i_o \exp\left(\frac{\alpha_a F \eta}{RT}\right)$$

$$\frac{i}{i_o} = \exp\left(\frac{\alpha_a F \eta}{RT}\right)$$

$$0.9 = \exp\left(\frac{(0.5)(96485)}{RT}\eta\right)$$

$$|\eta| = 5.32 \times 10^{-3}\ V$$

@ 99% of total current:

$$\frac{i}{i_0} = 0.99 = \exp\left(\frac{\alpha_a F \eta}{RT}\right)$$

$$|\eta| = 5.08 \times 10^{-2}\ V$$

It seems that we are in linearized B-V region i.e. use linear kinetics to model the system

Problem 8

① $Cu \rightleftharpoons Cu^+ + e^-$ (RDS) $\qquad i = f(V, C_k)$

② $Cu^+ + 2Cl^- \rightleftharpoons CuCl_2^-$

① Electrochem:

$$r_1 = k_{1,a}\exp\left(\frac{(1-\beta)FV}{RT}\right)[Cu] - k_{1,c}[Cu^+]\exp\left(\frac{-\beta FV}{RT}\right)$$

② Chemical:

$$r_2 = k_f[Cu^+][Cl^-]^2 - k_r[CuCl_2^-]$$

r_1 is rds, r_2 is QE: $\quad r_2 \approx 0 \Rightarrow \quad k_f[Cu^+][Cl^-]^2 = k_r[CuCl_2^-]$

$$[Cu^+] = \frac{k_r}{k_f}\frac{[CuCl_2^-]}{[Cl^-]^2}$$

$$r_{net} = r_1 = k_{1,a}[Cu]\exp\left(\frac{(1-\beta)FV}{RT}\right) - k_{1,c}[Cu^+]\exp\left[\frac{-\beta FV}{RT}\right]$$

$$\frac{i}{F} = k_{1,a}[Cu]\exp\left(\frac{(1-\beta)FV}{RT}\right) - \frac{k_{1,c}k_r}{k_f}\frac{[CuCl_2^-]}{[Cl^-]^2}\exp\left(\frac{-\beta FV}{RT}\right) \rightarrow \text{Matches B-V kinetics (w/ 1 } e^- \text{ transfer)}$$

Solving for exchange current: $i = 0$ when $V = U$

$$\therefore \quad k_{1,a}[Cu]\exp\left(\frac{(1-\beta)FU}{RT}\right) = \overbrace{\frac{k_{1,c}k_r}{k_f}\frac{[CuCl_2^-]}{[Cl^-]^2}}^{Z}\exp\left(\frac{-\beta FU}{RT}\right)$$

$$\frac{k_{1,a}[Cu]}{Z} = \exp\left(\frac{-\beta FU}{RT} - \frac{(1-\beta)FU}{RT}\right) \qquad -\beta - 1 + \beta$$

$$\frac{k_{1,a}[Cu]}{Z} = \exp\left(\frac{-FU}{RT}\right)$$

$$\frac{FU}{RT} = -\ln\left(\frac{k_{1,a}[Cu]}{Z}\right) \qquad \left[\exp\left(\frac{FU}{RT}\right) = \frac{Z}{k_{1,a}[Cu]}\right]$$

$$U = \frac{RT}{F}\ln\left(\frac{Z}{k_{1,a}[Cu]}\right)$$

$\eta = V - U$; $V = \eta + U \rightarrow$ Plugging in the original expression:

$$\frac{i}{F} = k_{1,a}[Cu]\exp\left(\frac{(1-\beta)FV}{RT}\right) - \frac{k_{1,c}k_r}{k_f}\frac{[CuCl_2^-]}{[Cl^-]^2}\exp\left(\frac{-\beta FV}{RT}\right)$$

$$= k_{1,a}[Cu]\exp\left(\frac{(1-\beta)F}{RT}(\eta+U)\right) - Z\exp\left(\frac{-\beta F}{RT}(\eta+U)\right)$$

$$= k_{1,a}[Cu]\exp\left(\frac{(1-\beta)F\eta}{RT}\right)\exp\left(\frac{(1-\beta)FU}{RT}\right) - Z\exp\left(\frac{-\beta F}{RT}\eta\right)\exp\left(\frac{-\beta F}{RT}U\right)$$

$$= k_{1,a}[Cu]\exp\left(\frac{(1-\beta)F\eta}{RT}\right)\exp\left(\frac{FU}{RT}\right)\exp\left(\frac{-\beta FU}{RT}\right) - Z\exp\left(\frac{-\beta FU}{RT}\right)\exp\left(\frac{-\beta F\eta}{RT}\right)$$

$$= \frac{k_{1,a}[Cu]}{k_{1,a}[Cu]}Z\cdot\exp\left(\frac{-\beta FU}{RT}\right)\left[\exp\left(\frac{(1-\beta)F\eta}{RT}\right) - \exp\left(\frac{-\beta F\eta}{RT}\right)\right]$$

$$= z \cdot \exp\left(-\frac{\beta \cdot F}{RT} \cdot \frac{RT}{F} \ln\left(\frac{z}{k_{1,a}[Cu]}\right)\right) \left[\quad '' \quad \right]$$

$$= z \cdot \exp(-\beta) \frac{z}{k_{1,a}[Cu]} \left[\quad '' \quad \right]$$

$$i = \underbrace{\frac{z^2 \cdot \exp(-\beta)}{k_{1,a}[Cu]}}_{i_0} \left[\exp\left(\frac{(1-\beta) F\eta}{RT}\right) - \exp\left(-\frac{\beta F \eta}{RT}\right) \right]$$

$$i_0 = \left[\frac{k_{1,c}\, k_r}{k_f} \frac{[Cu_2Cl_4^{-}]}{[Cl^-]^2}\right]^2 \cdot \frac{1}{k_{1,a}[Cu]} \exp(-\beta)$$

$$i_0 \propto \frac{[Cu_2Cl_4^{-}]^2}{[Cl^-]^4 [Cu]}$$